Z00510

SHORT QUESTION

Q. Broiler/Flyer:

Chickens grown for meat production

Q. Species cultured in ponds

- Rohu Labeo rohita
- Morakhi Cirrhinus mrigala
- Catla/Thaila Catla catla
- Grass carp Ctenopharyngodon idella
- Silver carp Hypophthalmichthys molitrix
- Big head carp Aristichthys nobilis
- Common carp Cyprinus carpio Tilapia Oreochromis niloticus

Q. Functions of fat:

1-Providing energy

2-Cushions for vital organs

3-Energy reserves

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4-Insulators & lubricants

5-Transports of fat-soluble vitamins

Q. White Spot Disease itch.

- Causes

Protozoan parasite either free swimming in the water or carried in with new fish or plants.

Fish under stress from bad water conditions are more susceptible.

- Symptoms

• The fish's skin and fins are covered in tiny white spots A badly affected fish may make rapid gill movements.

Q. Define Catching

- Once the birds reach their required weight they are transported to the processing plant.
- Trained staff called 'catchers' catch the birds and put them in special containers called modules.
- Catching is carried out quietly and with care to avoid unnecessary stress and to prevent injury to the birds

Q. Difference between queen and drone bee.

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The Queen

- The queen is the only sexually developed female in the hive.
- She is the largest bee in the colony.
- A productive queen can lay 2,000 eggs in a single day.

The Drone

- Drones are stout male bees with large eyes and no stingers.
- Drones do not collect food or pollen from flowers.
- Their sole purpose is to mate with the queen.
- They die upon mating.
- If the colony is short on food, drones are often kicked out of the hive.

Q. Factors affecting the requirements of protein

a) More Waste Limited Digestion

b) About 36% of feed is excreted as organic waste

c) Live fish biomass = 5x more waste than human

d) Why?

i. Limited digestion ii. Large fraction of feed remain

undigested & is excreted .

Q. Advantage and disadvantage of Hand feeding method

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- Feed is distributed by hand, shovels, or other non-automated means.
- Most common method.
- Used for feeds not suited for automated systems.
- Meat scraps are hand fed.

Advantages:

1-Operator can note feeding behavior gauge the feed required
2Operator can ensure that feed is dispersed over wide area.

Disadvantages:

1-high labor cost

2-increased handling of the feed.

Q. What is raceway

Raceways are considered flow through systems. Being simple to construct they are some of the oldest designs in aquaculture. Water sources for raceway aquaculture operations are usuallystreams, springs, reservoirs or deep wells.

Q. Apiary.

The place where a beekeeper keeps his bees is called an apiary or a bee yard.

Q. Difference between demand and anger feeder.

a. Demand feeders

- Allow fish to bump a rod and release feed when they are hungry.
- Used in raceways.
- Fish must be trained how to use them.

Fish can obtain food on demand by depressing a trigger.

• Auger feeding

• Similar to systems used by swine and itry farmers.

• Feed stored in large tanks

• Moved by augers that drop feed in pond

• Common in tanks and raceways

Q. Importance of lacwith increasing universal environment awareness

The importance of lac has assumed special relevance in the present age, being

a) An eco-friendly

b)Biodegradable

c) Self-sustaining natural material

Since lac insects are cultured on host trees which are growing primarily in wasteland areas, promotion of lac and its culture can help in eco-system development as well as reasonably high

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economic returns. It is a source of livelihood of tribal and poor inhabiting forest and subforest areas.

Q. Inoculation:

The method by which the lac insects are introduced to the new lac host plant is known as inoculation.

Q. Ari lac and Phunki lac

Lac can be scraped from the twigs before or after the emergence of larvae. If it is used for manufacturing before the emergence of larvae, the type of lac produced is called as “Ari lac”and

if it is used for manufacturing purpose after swarming of larvae has occurred, the lac is said to be “Phunki lac”.

Q.Bioaccumulation:

When a chemical accumulates in animal fat (historical fact) is called bioaccumulation.

Q.Animal by-Products:

The animal by products are as follows:

1-Bones =Button, glue, mineral supplement for livestock feed (Ca)

2--Fat =Chemicals, salves, creams, dressings, lubricants, soaps, food

3- Glands =Medicines, food additives

4-Collagen=Glue, Gelatin

5- Intestinal & Stomach tissue =lunch meats, surgical sutures, strings for musical and sports instruments

6- Fertilizer=Natural

Q.Bubonic Plague:

The plague caused by a bacterium (Yersinia Pestis) and characterized by especially by the formation of buboes is called bubonic plague. This Plague is currently managed monitoring for plague and reducing the number of rodenthosts for fleas.

Q.Average milk composition:

The average milk composition is as follows:

Milk:

a-Water 87.5% b-Total Milk Solids 12.5%

1-Fat 3.8%

2- Solids Non Fat (SNF) 8.7%

i-Lactose 4.6%

ii-Protein (Casien) 3.1% iii-N

Protein 0.2% iv-Vitamins &

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Minerals 0.8%

Plants pathogens:

Pathogens: Fungi – Bacteria – Viruses – Nematodes – Phytoplasmas are the pathogens of plants.

Q.Integrated Pest Management(IPM):

Implementation is done by:

1. Select control strategies

2. Timing

3. Economics

4. Environmental impacts 5. Regulatory restrictions

Why practice IPM:

The IPM practice for to Maintains balanced ecosystems, pesticides alone may be ineffective , promotes a healthy environment ,saves money and maintains a good public image

Q.How many exposures does it take to make your population resistant?

The exposure that make our population resistant are as follows: Toss your population (pennies) on the ground, and consider all with painted sides showing as exposed to pesticide:

1- All exposed yellow (RR = susceptible) die so remove these pennies.

2– All exposed red (rr = resistant) survive and double;

3-add a red penny for each exposed red penny.

4– For every three exposed blue (mild resistance) add one of each color. (Keep track of any remaining blues

Q. Hygiene and Milk Quality:

A science of the establishment and maintenance of health is called hygiene and white liquid with balance diet is called milk.

The hygiene and milk quality is as follows:

The milk without dirt (bacteria or antibiotics, poor milk filters by bacteria or antibiotics, low cream and animal mastitis) is hygiene and milk quality.

Q.Ecological pest management:

The management that focuses on prevent rather than reactive approaches to pest management by increasing biodiversity and creating habitat for natural enemies, land managers can boost natural pest control services and reliance on pesticides in ecology is the ecological pest management.

Q.Natural pest control factors:

The natural pest control factors are as follows:

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1-Wind

2-Temperature

3-Humidity, rain

4-Rivers, lakes, mountains

5- Pathogens, predators

6-Food supply of the pest

Q.Characteristics of Phylum arhropods:

1-Highly successful and segmented Eucoelomata with exoskeleton (Arthropod Metamerism)

2-Tagmata/Tagmatization (segments grouped into body regions)

3-Molting, Ecdysis (Ecdysozoa)

4- Hemimetabolous (gradual) / holometabolous (complete)

5- Cephalization (highly developed sensory organs)

6- Open circulatory system (hemolymph / hemocoel)

7- Respiratory system (trachea / book lungs)

8- Malpighian tubules / Coxal glands

9- Involved in virtually every kind of parasitic relationship

Q.Monogastric and Polygastric:

MONOGASTRIC:The animal having a simple stomach with only a single compartment is called monogastric. for example: Pig, Dog, Human .

POLYGASTRIC:The animal having a stomach with more than one compartment is called polygastric. For example: (ruminants) – Cow, Sheep, Goat.

Q.Legs segment in animals:

The jointed appendages are legs segment of animals especially in arthropods. The legs segment of animals that provide travel and protect sensitive organ from damages.

Q.Housefly mouth pars:

Fulcium, clypeus, rostrum, stipes, haustellum,theca, maxillarypalp, labrum, prementum, labellum, pseudotracea, furca and prestomal teeth are mouth parts of housefly.

Q.Vector-Borne Diseases:

diseases : Vectors

1-Dangue: Mosquito

2-Typhus: Rodent

3-Asthma: Cockroach

4-Bubonic Plague: Fleas

5-Skin infection: Louse

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Q.Insect life cycle:

1- Egg ----2-larvae----3-pupa----4-adult---

Q.Hemimatabolous and holometabolous:

The gradual or incomplete metamorphosis is called hemimetabolous and complete metamorphous is called holometabolous.

Q.How to control weeds:

Weeds are control by cultural method by using mowing ,grazing, over seeding and improved fertility.

Q.Sources of proteins:

Animals and plants are sources of proteins.

Q.Vector biting Activities:

The nocturnal , diurnal and crepuscular are vector biting activities.

Q. Technique of lac culture?

Lac is a natural resin of animal origin. It is secreted by an insect, known as lac-insect in order to obtain lac, these insects are cultured and the technique is called lac-culture.

Q. Protein requirements for fish?

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Fish contain 60-93% crude protein, thus, fish diets are higher in protein than birds & animals.

Fish appear to be relatively efficient utilizer of protein to energy with 84%.

Fish can not synthesize their amino acids & obtain chiefly from the diet & absence affects growth of fish

Q.Composition of lac.

Lac is a mixture of several substances, of which resin is the main constituent. The approximate percentage of different constituents of lac is given below:

Resin – 68 to 90%

Dye – 2 to 10%

Wax – 5 to 6%

Mineral matter – 3 to 7%

Albuminous matter – 5 to 10%

Water – 2 to 3%

Q. Spring viremia crops.

A viral infection caused by Rhabdovirus carpio.

Symptoms

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Darkening of skin, pale gills, pop eye, protruding vent, bleeding in gills skin and eyes. Lethargy, abnormal swimming positions, sitting on bottom of the tank.

Treatment:

No known treatment .

Q. Guard bees and soldier bees.

Guard Bees - protect the entrance of the hive from enemies.

Soldier bees - Soldiers hang around near the entrance and attack invaders. They work in concert

Q. Plumage.

Plumage: Feathers of a bird.

Cause :with entrance guards.

Q. What is virus?

A virus is a microscopic organism that can only reproduce by inhabiting host cells and using the genetic material in the cells of a host.

Q. What is hatchery?

The eggs are incubated at the hatchery. They are kept warm, until the chicks start to hatch out of their shells. Chickens hatch at around 20 days and turkeys hatch at around 27 days. Baby chickens are called chicks. Baby turkeys are called poults.

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Q. Oligotrophic & eutrophic.

Oligotrophic

- Excellent water quality
- Very Clear
- Used in display aquaria
- Most frequently used for breeding purposes
- Some species are kept in these conditions all of their lives, while others for a period of time

Eutrophic

Exist for the grow out of the most tolerant species that show vigorous growth under moderately deteriorated water quality conditions

Dissolved oxygen levels- economic optimum level

Ammonia & Nitrite – less than 2mg-N/L

Water quality – marginal

Species evolved under similar natural conditions prosper in these conditions

Q. What is difference between intensive and extensive aquaculture.

Extensive Aquaculture: Minimal control, lower density • E.g., ponds, prevalent in third world countries

Intensive Aquaculture: Highly controlled, high density, RAS, raceways, confined • E.g. industrialized .

Q. Ecological pest management.

Ecological Pest Management (EPM) is an approach to increasing the strengths of natural systems to reinforce the natural processes of pest regulation and improve agricultural production.

1. Control based on pest life cycle and ecology

2. Control agent may be an organism or chemical

3. Specific to pest and/or manipulate a part of the ecosystem

4. Emphasizes protection from pest

Q. What is “Biosecurity”?

Protecting your birds from disease .Preventing or controlling disease transmission by vectors

• procedures or measures designed to protect the population against harmful biological or biochemical substances.

Q. What is a vector?

• Something that may transmit a disease

• Rodents, birds, insects, shoes, car tires, shared equipment, best friend, pet, feed, water, dust, air….

A VECTOR is an organism that does not cause disease itself but which spreads infection by conveying pathogens from one host to another. Species of mosquito, for example, serve as vectors for the deadly disease Malaria.

Q. How many exposures does it take to make your population resistant?

Expose a population to a pesticide several times with mating in between exposures and see how many exposures it takes for resistant bugs to become the majority.

• Rules of the Game:

1. Toss your population (pennies) on the ground, and consider all with painted sides showing as exposed to pesticide

2. All exposed yellow (RR = susceptible) die so remove these pennies.

3. All exposed red (rr = resistant) survive and double; add a red penny for each exposed red penny.

4. For every three exposed blue (mild resistance) add one of each color. (Keep track of any remaining blues (<3) to ad to those on the next throw)

5. Keep track of the exposure count and repeat until most pennies are red.

THANK YOU


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